

TBA Delta Hedging

Generated: 2026-08-18 00:15:15 UTC

Model: claude-opus-5 · Effort: high · Max tokens: 8000

Question

You went Long **\$100 million face value** of a 10-Year Fannie Mae 5.5% TBA contract at a market price of \$99.34375. Your trade settles on the upcoming front-month **Settlement Date of August 13, 2026**. Assume a constant prepayment speed of **6% CPR**.

- 1. How much value does a portfolio lose when interest rates go up by one basis point?
- 2. You want to create a duration-neutral macro hedge using a combination of 5-Year and 10-Year Pay-Fixed SOFR Swaps. Your risk mandate requires exactly 40% of the total hedge DV01 to come from the 5-Year Swap and 60% from the 10-Year Swap. Assume the following swap DV01s:

5-Year Pay-Fixed SOFR Swap DV01:0.0455% of notional per 1 bp

10-Year Pay-Fixed SOFR Swap DV01: 0.0825% of notional per 1 bp

Using the TBA DV01 calculated in Part 1, Calculate the exact Notional Amounts of the 5-Year and 10-Year Swaps you must execute to create a perfect hedge.

G6					
	A	B	C	D	
1	Input	Value			
2	Face Value	\$100.00M			
3	TBA Price	99.34375			
4	Annual Coupon Rate	5.50%			
5	Remaining Term in Years	10			
6	Annual CPR	6%			
7	Months Remaining	120			
8	Initial Market Value	\$99.34M			
9	Monthly Coupon Rate	0.4583%			
10	Monthly SMM	0.5143%			
11	Monthly Payment	\$1.0853M	1 bps Jump	DV01	
12	Discount Factor	5.70%	5.71%	\$0.0326M	
13					

F20				
	A	B	C	D
1	Face Value	\$100.00M		
2	TBA DV01	\$0.0326M		
3				
4	Properties ▼	5Y Swap ▼	10Y Swap ▼	
5	Weight	40%	60%	
6	Target Hedge DV01	\$0.0130M	\$0.0195M	
7	Swap DV01 (% of Notional per 1 bp)	0.0455%	0.0825%	
8	Hedge Notional	\$28.6417M	\$23.6945M	
9	Hedge DV01 Generated	\$0.0130M	\$0.0195M	
10	Total Hedge DV01	\$0.0326M		
11				
12				
13	Net DV01 After Hedge	\$0.0000M		
14	Solver Error	\$0.0000M		
15				
16				
17				
18				
19				

Experiment setup

	Naive Agent	Guided Agent
Corpus access	corpus	additional_corpus , corpus
System prompt	none	Principal Fixed Income Quant Researcher + 5 guardrails
Tools	list_files , read_file , run_python	list_files , read_file , run_python

Each agent ran inside its own sandbox directory containing copies of only the corpora it was permitted to read. `additional_corpus/` — which holds the MBS cash-flow engine, the DV01 definition and the `[[Data Fallback]]` protocol — was absent from the Naive Agent's sandbox, so it had no path to those files.

Answer — Naive Agent

Part 1 Recap — The TBA DV01

Cash flow model: FNMA 5.5% 10-year pass-through, 120 months, TBA = new production (WALA 0) settling 8/13/2026, level-pay amortization at the 5.5% coupon, overlaid with a constant 6% CPR → **SMM = 0.51430%** ($1 - (1 - 0.06)^{(1/12)}$).

Metric	Value
SMM (6% CPR)	0.514301%
WAL @ 6% CPR	4.49 years
Price	99-11 (99.34375)
Implied cash-flow yield	5.6738%
Modified duration	3.783
TBA DV01	0.03758 pts = 0.0376% of face
DV01 on \$100mm face	≈ \$37,577 per basis point

Part 2 — Hedge Notionals

You are **long** the TBA (long duration), so you **pay fixed** on both swaps to offset. Split the \$37,577 DV01 40/60:

Leg	Target DV01	Swap DV01	Notional to Execute
5-Yr Pay-Fixed SOFR	40% × \$37,577 = \$15,030.78	0.0455% (=\$455/\$1mm)	\$33,034,677 (≈ \$33mm)
10-Yr Pay-Fixed SOFR	60% × \$37,577 = \$22,546.17	0.0825% (=\$825/\$1mm)	\$27,328,687 (≈ \$27mm)
Total	\$37,576.94	—	\$60,363,364

The arithmetic:

- 5Y: \$15,030.78 ÷ 0.000455 = **\$33.03mm**
- 10Y: \$22,546.17 ÷ 0.000825 = **\$27.33mm**

Total hedge notional is only **0.60× the face** of the TBA — the amortizing, prepaying MBS has a much shorter duration (3.78) than a bullet swap, so far less notional is needed.

Verification (parallel shocks): net P&L is $\pm \$1.1\text{k}$ at 10bp and $\pm \$27\text{k}$ at 50bp on a \$100mm position — DV01 is neutralized, residual is pure second-order.

Critical Caveats

1. **This is a static (constant-CPR) DV01, not an effective DV01.** Holding CPR fixed at 6% forces the model to show *positive* convexity (+21.9). Real MBS are **negatively convex** — rates rally, prepays accelerate, duration shortens. A true OAS-based effective duration would be materially lower, and the hedge would be **over-hedged in a rally, under-hedged in a sell-off**. You must re-balance dynamically.
2. **The 40/60 mandate is a risk-limit constraint, not a curve hedge.** It is imposed by mandate, not derived from the TBA's key-rate profile. With a 4.49-year WAL, the bucketed exposure sits far closer to the 5-year point; the 60% weight on the 10-year leaves you structurally **exposed to 5s/10s curve steepening**.
3. **Basis risk remains.** Swap DV01s hedge SOFR rates only — mortgage spread/OAS widening, and the fact that TBA yields off the delivery option (cheapest-to-deliver) rather than a specific pool, are unhedged.
4. **Assumptions to confirm:** I modeled "10-Year" as a 120-month pass-through with no servicing/g-fee gap (WAC = net coupon) and no payment delay. Adding Fannie's 24-day delay or a WAC above 5.5% would shift the DV01 modestly; if "10-Year" instead means a 30-year pool benchmarked to the 10-year point, the DV01 roughly doubles and so do all hedge notionals.

Answer — Guided Agent

Part 1 Recap — TBA DV01

Deterministic (fixed-cashflow) pipeline per the corpus, 120-month schedule at 6% CPR:

Item	Value
Face Value	\$100,000,000
Market Value (100mm $\times$ 99.34375/100)	\$99,343,750.00
SMM (from 6% CPR)	0.514%
Market-Implied Discount Rate (solved)	5.700426%
PV_{current}	\$99,343,750.00
$PV_{+1\text{bp}}$	\$99,311,170.04
TBA DV01 (long position)	\$32,579.96 per bp

The long TBA has **+\$32,579.96** of DV01 to neutralize, so the hedge must generate an offsetting **−\$32,579.96** via **pay-fixed** swaps (pay-fixed = negative delta per the corpus convention).

Part 2 — Hedge Risk Allocation

$$\Delta_1^{target} = 0.40 \times 32,579.96 = \$13,031.98 \quad \Delta_2^{target} = 0.60 \times 32,579.96 = \$19,547.97$$

Part 3 — Solved Notionals

$N_1 = \frac{w_1 \Delta_{target}}{d_1}$, $N_2 = \frac{w_2 \Delta_{target}}{d_2}$ — closed form cross-validated against the SciPy least-squares solver (error $\approx 1.7\text{e-}23$).

Instrument	DV01 / \$1 notional	Required DV01	Notional to Execute
5-Yr Pay-Fixed SOFR Swap	0.000455 (0.0455%)	\$13,031.98 (40%)	\$28,641,720.57 ($\approx$ \$28.64mm)
10-Yr Pay-Fixed SOFR Swap	0.000825 (0.0825%)	\$19,547.97 (60%)	\$23,694,514.29 ($\approx$ \$23.69mm)
Total	—	\$32,579.96	\$52,336,234.86

Acceptance Criteria — All Passed

- Δ_1 generated \$13,031.98 $\approx$ Δ_1 target ✓
- Δ_2 generated \$19,547.97 $\approx$ Δ_2 target ✓
- Δ_{hedge} \$32,579.96 $\approx$ Δ_{target} ✓
- **$\Delta_{\text{net}} = \$0.0000$** ✓ (duration-neutral)
- Solver Error ≈ 0 ✓

Execution ticket: Pay fixed on **\$28.64mm 5-Year SOFR** and **\$23.69mm 10-Year SOFR** (total \$52.34mm notional = 52.34% of TBA face, reflecting the short ~3.77-year WAL of a 6% CPR 10-year 5.5% pass-through).

⚠ Caveat: This is a *static/fixed-cashflow* DV01 — cash flows are held constant at 6% CPR under the rate shock, so it captures no negative convexity. Real-world effective duration will shorten as rates rally (prepay burnout reversal), meaning this hedge will become **over-hedged in a rally and under-hedged in a sell-off**; it requires dynamic rebalancing or an OAS/Monte Carlo effective-duration overlay.

Run metadata

Field	Naive run	Guided run
Model	claude-opus-5	claude-opus-5
Effort	high	high
First request ID	req_011Ce9FH53bKCjFBotfw8RJX	req_011Ce9FPgCrSRtcjpyNh18G6
Final response ID	msg_011Ce9FN7tdsRGXqF7dLsgD1	msg_011Ce9FSArqn89BAaYagoeDb
Start (UTC)	2026-08-18T00:12:56.277072+00:00	2026-08-18T00:14:25.984824+00:00
End (UTC)	2026-08-18T00:14:25.974682+00:00	2026-08-18T00:15:15.456806+00:00
Wall clock (s)	89.698	49.478
API calls	7	6
Tool calls	8	5
Input tokens	110,322	59,803
Output tokens	6,537	3,819
Cache read tokens	0	0
Cache creation tokens	0	0
Final stop reason	end_turn	end_turn

API call trace — Naive Agent

#	Request ID	Response ID	Stop	In
1	req_011Ce9FH53bKCjFBotfw8RJX	msg_011Ce9FH66M6AegQPEokvLzR	tool_use	95
2	req_011Ce9FHHKn8FXicYXi9BsTx	msg_011Ce9FHHyU2CxSuzSuhRbWi	tool_use	16
3	req_011Ce9FHVslWMUjE9zpZcvK1	msg_011Ce9FHWYky7ifhNttKi2sb	tool_use	18
4	req_011Ce9FKMS8v1vbMdfVagSaG	msg_011Ce9FKNBnBg5TWT5yMiNKU	tool_use	20
5	req_011Ce9FL73yeQZtoyWTi8BMq	msg_011Ce9FL7r6ZtxeHYrf1wn7v	tool_use	21

#	Request ID	Response ID	Stop	In
6	req_011Ce9FMMVQrhU7NuA9hC6Pn	msg_011Ce9FMN9bpeNr5FC4AxPtY	tool_use	23
7	req_011Ce9FN7DTCvNh9kwnbhN3x	msg_011Ce9FN7tdsRGXqF7dLsgD1	end_turn	24

API call trace — Guided Agent

#	Request ID	Response ID	Stop	In
1	req_011Ce9FPgCrSRtcjpyNh18G6	msg_011Ce9FPgu2RssezAoudhRB3	tool_use	12
2	req_011Ce9FPogp7gF5JGSZggARm	msg_011Ce9FPpMFTMCtk8vK91ce4	tool_use	18
3	req_011Ce9FPygKyBDCGB8ZuUvnm	msg_011Ce9FPzQVCJ9WnUCm62kJ4	tool_use	11
4	req_011Ce9FQFpgXNo5rajXcc6Qj	msg_011Ce9FQGcoipGWcDMNWP63F	tool_use	13
5	req_011Ce9FRWRnqYkwi5WbtkZyE	msg_011Ce9FRXGtWdmNagvUgfHy1	tool_use	15
6	req_011Ce9FS9x22mJbVyVfbZrCN	msg_011Ce9FSArqn89BAaYagoeDb	end_turn	16

Tool call trace — Naive Agent

#	Tool	Input (truncated)	Error
1	list_files	{}	no
1	run_python	{"code": "print(\"env ok\")"}	no
2	read_file	{"path": "corpus/data/md_outputs/21passthh.md"}	no
2	read_file	{"path": "corpus/data/md_outputs/securitization.md"}	no
3	run_python	{"code": "\nimport numpy as np\nfrom scipy.optimize import brentq\n\nface = 100_000_000\nprice = 99.34375"}	no
4	run_python	{"code": "\nimport numpy as np\nfrom scipy.optimize import brentq\n\nexec(open('/dev/null').read()) if False else"}	no
5	run_python	{"code": "\nimport numpy as np\nfrom scipy.optimize import brentq\n\nface=100_000_000; price=99.34375; cpn=.055;"}	no

#	Tool	Input (truncated)	Error
6	run_python	{"code": "\nimport numpy as np\nfrom scipy.optimize import brentq\nface=100_000_000; price=99.34375; n=120; sm	no

Tool call trace — Guided Agent

#	Tool	Input (truncated)	Error
1	list_files	{}	no
2	read_file	{"path": "additional_corpus/tba_delta.md"}	no
3	read_file	{"path": "additional_corpus/credit_card_securitization.md"}	no
4	run_python	{"code": "\nimport pandas as pd\nfrom scipy.optimize import brentq\n\ndef pmt(rate, nper, pv, fv=0.0, payment_	no
5	run_python	{"code": "\nfrom scipy.optimize import minimize\nimport numpy as np\n\ndv01_target = 32579.957148700953 # fr	no

Credit Card Securitization

Generated: 2026-08-18 19:57:06 UTC

Model: claude-opus-5 · **Effort:** high · **Max tokens:** 8000

Question

An SPV purchases a portfolio of credit card receivables with an initial balance of \$500 million. The portfolio contains 400,000 customer accounts. The average annual percentage rate (APR) charged on the receivables is 18%. The SPV pays an annual servicing fee equal to 2% of the beginning receivable balance. The portfolio has an annual charge-off rate of 5%. Recoveries equal 20% of charged-off balances, and recoveries are received after a 4-month recovery lag. The model has a 60-month horizon.

During months 1 through 12, customers pay 10% of their outstanding balance each month. During months 13 through 24, customers pay 12% of their outstanding balance each month. From month 25 onward, customers pay 15% of their outstanding balance each month.

The transaction has an 8-month revolving period. During the revolving period, new receivables are purchased each month. New purchases equal 8% of the beginning receivable balance. After the revolving period ends, no additional receivables are purchased. The SPV finances the receivables by issuing the following securities: Class A Notes: \$375 million, Class B Notes: \$75 million, Seller's Interest: \$50 million. Class A Notes pay a floating coupon equal to SOFR plus 1.0%. Class B Notes pay a floating coupon equal to SOFR plus 3.0%. SOFR is assumed to be 4%. During the revolving period, principal collections are used to support new receivable purchases, and no principal payments are made to Class A or Class B. After the revolving period ends, principal collections and recoveries become available for repayment of the notes. The available principal is distributed pro rata between Class A and Class B based on their original principal balances. Interest payments are made to Class A and Class B based on their respective outstanding principal balances and contractual coupon rates. The remaining cash after the required payments to the noteholders is allocated to the Seller's Interest. Finally, calculate the total amount of payments received by Class A and Class B over the 60-month modeling horizon.

	A	B	C	D	E	F	G	H	I
1	Explanation	Parameter	Value						
2	Total outstanding credit card debt purchased by the SPV. This is the asset backing the ABS.	Initial Receivable Balance	\$500 M						
3	Number of individual credit card customers. Mainly informational in this model.	Number of Accounts	400000						
4	Interest charged by the bank on outstanding credit card balances. This is the income generated by the receivable pool.	Annual APR	18%			Tranche	Notes	SOFR	Spread
5	Fee paid to the bank (or servicer) for managing the accounts, collecting payments, handling defaults, etc.	Annual Servicing Fee	2%			Class A	\$375 M	4%	1%
6	Percentage of receivables expected to become uncollectible each year. A charge-off removes the receivable from the books.	Annual Charge-off Rate	5%			Class B	\$75 M		3%
7	Percentage of charged-off balances that are eventually recovered through collections or legal action.	Recovery Rate	20%			Seller's Interest	\$50 M		
8	Number of months after a charge-off before the recovery cash is received.	Recovery Lag (Months)	4						
9	Number of months during which principal collections are used to buy new receivables instead of repaying investors.	Revolving Period	8						
10	New credit card spending each month as a percentage of the beginning receivable balance.	New Purchase Rate	8%						
11	Total number of months to project.	Model Horizon	60			Class A Total Payment	\$397.9383 M		
12						Class B Total Payment	\$88.4904 M		
13						Total	\$486.4287 M		
14		Months	MPR						
15		1–12	10%						
16		13–24	12%						
17		25–60	15%						

Experiment setup

	Naive Agent	Guided Agent
Corpus access	corpus	additional_corpus , corpus
System prompt	<i>none</i>	Principal Fixed Income Quant Researcher + 5 guardrails
Tools	list_files , read_file , run_python	list_files , read_file , run_python

Each agent ran inside its own sandbox directory containing copies of only the corpora it was permitted to read. `additional_corpus/` — which holds the MBS cash-flow engine, the DV01 definition and the `[[Data Fallback]]` protocol — was absent from the Naive Agent's sandbox, so it had no path to those files.

Answer — Naive Agent

Model Setup

Pool mechanics (monthly, on beginning balance): finance charges = APR/12, servicing = 2%/12, charge-offs = 5%/12, principal collections = MPR × balance, recoveries = 20% × charge-offs lagged 4 months, purchases = 8% × balance for months 1–8.

Ending Bal = Beginning – Principal Collections – Charge-offs + New Purchases

Liabilities: Class A \$375M @ 5.0% (SOFR 4% + 1%), Class B \$75M @ 7.0% (SOFR + 3%), Seller \$50M. Principal after month 8 split 83.33% / 16.67% (pro rata on original balances).

Collateral Behaviour

	Month 1	Month 8 (end revolving)	Month 12	Month 24	Month 60
Beginning balance	\$500.0M	\$421.3M	\$295.6M	\$84.9M	~\$0.15M
MPR	10%	10%	10%	12%	15%
Principal collected	\$50.0M	\$42.1M	\$29.6M	\$10.2M	~\$0.02M

During the 8-month revolving period the pool declines only slowly (10% MPR vs. 8% reinvestment), then amortises rapidly once purchases stop.

60-month pool totals: finance charges \$106.9M · principal collections \$764.4M · recoveries \$5.9M · charge-offs \$29.7M · servicing fees \$11.9M · new purchases \$294.2M

Note Cash Flows

Item	Class A	Class B	Total
Interest paid	\$33.07M	\$9.26M	\$42.33M
Principal paid	\$334.12M	\$66.82M	\$400.95M
Total received	\$367.19M	\$76.08M	\$443.27M
Balance still outstanding at M60	\$40.88M	\$8.18M	\$49.06M

Selected months (interest / principal):

- M1–M8 (revolving): A \$1.563M interest only; B \$0.438M interest only — no principal
- M9: A \$1.563M + \$34.58M; B \$0.438M + \$6.92M
- M13: A \$1.070M + \$26.76M; B \$0.299M + \$5.35M
- M25: A \$0.354M + \$6.81M; B \$0.099M + \$1.36M
- M60: A \$0.170M + \$0.019M; B \$0.048M + \$0.004M

Seller's Interest receives \$127.89M of residual cash (large early flows of ~\$12–15M/month during the revolving period, since 10% principal collections exceed the 8% reinvestment; declining and slightly negative after ~month 30 as excess spread thins).

Answer

Total payments to Class A and Class B over 60 months = \$443.27 million

- Class A: **\$367.19M** (\$334.12M principal + \$33.07M interest)
- Class B: **\$76.08M** (\$66.82M principal + \$9.26M interest)

Important caveat: the notes do **not** fully amortise in the 60-month window — \$49.1M remains outstanding (\$40.9M A, \$8.2M B) while the pool has run down to essentially zero. The gap arises because the 5% charge-off rate (with only 20% recovery, i.e. ~4% annual net loss) destroys collateral that pro-rata principal allocation never replaces; the geometric amortisation profile also leaves a long tail of tiny payments. In a real structure this shortfall would be covered by excess spread trapped in a reserve account (the ~\$128M of residual cash shown above is more than sufficient), rather than being left as a principal loss.

Answer — Guided Agent

Credit Card ABS — 60-Month Cash Flow Model Results

Protocol Note on Principal Allocation

Your prompt specified **pro-rata** principal distribution between Class A and Class B. However, the governing `additional_corpus/credit_card_securitization.md` protocol explicitly states:

Rule 5: Class A principal is paid before Class B principal. **Rule 6:** Do **not** use pro-rata principal allocation.

Per your directive to give the corpus greater weight, the model was run **sequentially (fully turbo)**: Class A absorbs all available principal until retired, then Class B.

Key Model Parameters (derived)

Item	Value
Monthly APR	1.5000%
Monthly servicing fee	0.16667%
Monthly charge-off rate	0.41667%
Class A coupon (SOFR 4% + 1%)	5.0% / yr → 0.41667% mo

Item	Value
Class B coupon (SOFR 4% + 3%)	7.0% / yr → 0.58333% mo
Revolving period	Months 1–8 (available principal = 0)
Recovery lag	4 months, 20% of charge-offs

Structural Dynamics Observed

- **Months 1–8 (Revolving):** Receivables decline gently from \$500.0M to \$468.8M — the 8% monthly repurchase nearly, but not fully, offsets the 8.5% net principal + charge-off drain. Notes remain at full \$375M / \$75M; interest is paid in full.
- **Month 9 (Amortization begins):** \$39.88M of principal + recoveries hits Class A immediately.
- **Month 13 & 25:** Payment-rate step-ups (10% → 12% → 15%) accelerate collections, but the shrinking receivable base dominates.
- **Month 25:** Class A fully retires (\$1.87M final paydown); Class B begins amortizing with the \$9.01M residual.
- **Month 51:** Class B fully retires (\$167.7K final paydown). Months 52–60 collections flow entirely to Seller's Interest.
- **Interest coverage:** Available interest (interest income – servicing fee) covers both classes in full every month — no shortfalls. Note in Month 25, Class A's interest due drops to \$7,809 on the \$1.87M stub balance, correctly computed off the **beginning** balance per Rules 1–2.

FINAL ANSWER — Total Payments Received (60-Month Horizon)

Class	Interest Paid	Principal Paid	Total Payments
Class A Notes	\$22,938,312	\$375,000,000	\$397,938,312
Class B Notes	\$13,490,391	\$75,000,000	\$88,490,391
Combined A + B	\$36,428,703	\$450,000,000	\$486,428,703

Both tranches are paid in full within the horizon — Class A retires in Month 25, Class B in Month 51, leaving \$362,556 of residual receivables and all excess spread flowing to the \$50M Seller's Interest.

Run metadata

Field	Naive run	Guided run
Model	claude-opus-5	claude-opus-5
Effort	high	high
First request ID	req_011CeAoTuRgpCYgKbs2qR5hq	req_011CeAoXtZxJHCbWie9tmLor
Final response ID	msg_011CeAoWaKKf6AC4taoxfuhk	msg_011CeAoZNVW9UxefQ2xt3wLB
Start (UTC)	2026-08-18T19:55:37.188841+00:00	2026-08-18T19:56:31.301824+00:00
End (UTC)	2026-08-18T19:56:31.283585+00:00	2026-08-18T19:57:06.612687+00:00
Wall clock (s)	54.095	35.311
API calls	3	4
Tool calls	2	4
Input tokens	12,870	15,462
Output tokens	4,355	2,749
Cache read tokens	0	0
Cache creation tokens	0	0
Final stop reason	end_turn	end_turn

API call trace — Naive Agent

#	Request ID	Response ID	Stop	In
1	req_011CeAoTuRgpCYgKbs2qR5hq	msg_011CeAoTvFMDtRGtPWB3RvZZ	tool_use	12
2	req_011CeAoV8cU2Wj fTBLVea7wW	msg_011CeAoV9KdJBxGDgX59uCQn	tool_use	42
3	req_011CeAoWZZg3J9fkY1MY1ddH	msg_011CeAoWaKKf6AC4taoxfuhk	end_turn	73

API call trace — Guided Agent

#	Request ID	Response ID	Stop	In
1	req_011CeAoXtZxJHCbWie9tmLor	msg_011CeAoXuD9h9KGFHFRakqq3	tool_use	15
2	req_011CeAoY7te5XnrP9s8bDwbM	msg_011CeAoY8b4h9DTQsBMjTC6M	tool_use	22
3	req_011CeAoYEtKo6pyZ25PCnMHo	msg_011CeAoYFeUKXoPrCzn6VnkF	tool_use	37
4	req_011CeAoZMdQwS56oxw1LafD4	msg_011CeAoZNVW9UxefQ2xt3wLB	end_turn	78

Tool call trace — Naive Agent

#	Tool	Input (truncated)	Error
1	run_python	{"code": "\nimport pandas as pd\nB=500e6\nAPR=.18;\nserv=.02; co=.05; rec=.20; lag=4\nrevolv=8;\npurch=.08\nA=37	no
2	run_python	{"code": "\nimport pandas as pd\nB=500e6; co_hist={}; rows=[]\nfor t in range(1,61):\n beg=B; pr=.10\nif t<=	no

Tool call trace — Guided Agent

#	Tool	Input (truncated)
1	list_files	{}
1	read_file	{"path": "additional_corpus.md"}
2	read_file	{"path": "additional_corpus/credit_card_securitization.md"}
3	run_python	{"code": "\ninitial_receivable=500e6\nannual_apr=0.18\nannual_servici